
Individual learning styles and design performance in the metaphorical reasoning process

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Abstract: Metaphor plays a key role in design practice. By framing problems in particular ways, metaphors not only impose structure on a design, but also determine the interpretations that can be made and approaches that can be taken. With the metaphorical reasoning process also it requires a wide range of skills essential for design, metaphors are suggested as a key area for investigating competencies that students may lack and therefore, as a means of helping instructors direct support for individuals in the design studio. Kolb's experiential learning theory, and the notion of learning styles, is proposed as a means of exploring the role of these competencies in different phases of metaphorical reasoning and for different kinds of individual learning tendencies. These relationships are then stated as hypotheses to be used in further investigations. Finally, recommendations for design education are suggested.

Keywords: accommodator; assimilator; converger; design education; design problem solving; diverger; experiential learning theory; learning styles; metaphorical reasoning.

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1 Introduction

Metaphors are acknowledged as being integral to architecture and product design (Schon, 1983a). They are used by designers to structure their approach to a given problem, allowing them to set boundaries and identify the potential relationships to be made (Coyne and Snodgrass, 1995). In the educational design studio however, this process is not well understood – indeed, with the exception of a few empirical investigations (Casakin, 2004a, 2005, 2006), the use of metaphors by design students has received little attention.

Different learning styles have also been shown to influence design problem solving. Demirbas and Demirkan (2003) for example, used Kolb's experiential learning theory (ELT; Kolb, 1984) to study the effects of learning style preferences on student performance during the design process and found statistically significant differences between the performance scores of students with diverse learning styles at different stages of the design process. These investigators also noted that the scores of students with different learning styles increased at the end of the design process.

In another study of architectural students in China (also using Kolb's theory), Kvan and Yunyan (2005) found a significant correlation between learning styles and academic performance that they went so far as to suggest that studio based programmes can actually disadvantage students with particular learning styles. In yet a further study, Bar-Eli (2005) used Kolb's model to identify the relationship between designers' learning styles, their characteristics and the designing behaviours observed during the design process – her aim being to provide confidence and support to students by helping them become aware of their learning strengths. She showed that the combination of individual designing characteristics is a determinant of designing behaviours and moreover, that such behaviours are intimately related to particular learning styles.

No study has considered the relationship between learning styles and the different phases of metaphor use in the design studio, yet there are potentially significant

advantages to be gained, for design studio teaching, from a better understanding of such a relationship – not least an increased focus on individuals and their unique learning tendencies. Gaining insight in learning styles could help us to better understand the individual learner and thus, provide one way of focusing instructors' efforts at areas in particular need of support. A better understanding of the relationship between metaphors and individual learning styles can also contribute to support and enhance the interactive and reflective dialogue established between the student and the design teacher in the different stages of the metaphorical process. A result of these may be an incremental development of design solutions along the design process, a consequence of which may be the production of more creative and innovative design outcomes. A theoretical study, focusing on architectural design students and the way they use metaphor is, therefore, discussed here as a basis for a future empirical study and as a means of opening a wider discussion in this important area. Kolb's ELT (Kolb, 1984) is taken as a framework for this investigation.

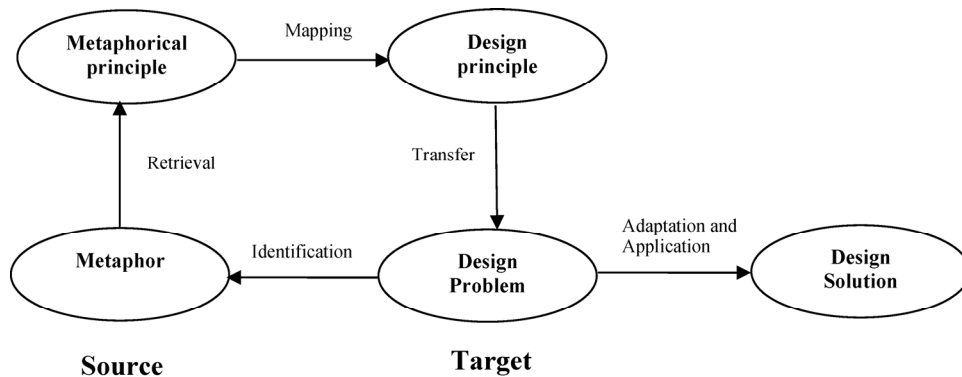
The first part of the discussion introduces the notion of metaphorical thinking. The component processes of this thinking are described and the role of metaphor in design problem solving (particularly in the design studio) is presented. In the second part of the article, Kolb's ELT is introduced and four kinds of learning styles are described and analysed with respect to metaphor use in design practice. A series of hypotheses about the effect of the different learning styles on each phase of the metaphorical process are then proposed before the main conclusions and potential implications for design education are presented.

2 Metaphorical thinking

Metaphor may be thought of as a comparison between two concepts that share certain characteristics, but differ in others. As such, they can be used to help us understand and reflect on one problem in terms of another that is not directly associated to it (Lakoff, 1987, 1993; Ortony, 1991). In language, a metaphor can be identified by the preposition 'as' or the verb 'is' (e.g. 'a house *is* a city' or 'a square *as* a temple of joy') and are ubiquitous in literature.

Metaphors are not, however, merely decorative – they also structure and organise our thoughts (Ricoeur, 1977). As Coyne and Snodgrass (1995) have claimed, metaphors are part of the study of rhetoric, concerned with how we argue, understand and solve a problem. Indeed, the ability of metaphor to help us define and elaborate problems from fresh viewpoints is considered to be pivotal to our capacity for problem solving (Schon, 1979) – and this is its subtle power. While at first glance a metaphor can seem to lack sense, further, careful, consideration can provide new insights by exposing aspects of a situation that would otherwise have been missed.

In essence then, metaphors allow for the transfer of concepts and ideas from remote domains to the problem at hand. An appropriate use of this mechanism 'temporarily removes the question of whether problems exist independently of our understanding of a domain – in other words, whether an 'objective' problem statement exists' (Coyne and Snodgrass, 1995).

Figure 1 Basic processes of metaphorical reasoning in design problem solving

2.1 Component processes in metaphorical reasoning

Since their basic processes are similar to those used in analogical reasoning, metaphors can be understood as a kind of analogy. Indeed, the structure-mapping theory developed by Gentner et al. (2001), unifies metaphor with processes of analogy and provides a set of psychologically tested processes that can thus be used to support the understanding of metaphor. By allowing metaphorical and analogical processes to be perceived within a single mechanism, structure-mapping theory enables metaphorical mapping to be understood as a process of establishing higher-order or structural alignment between two different situations with inferences being projected through a mapping of relationships (Gentner, 1983).

In this way, the mapping presupposes the emergence of an unambiguous set of correspondences that bear structured representations of objects and their properties, relations between objects, and higher-order relations between relations. The structure-mapping theory therefore suggests that metaphors can be processed as structural alignments that hinge on initial relational commonalities between source and target.

According to Gentner (1983), the basic processes of metaphorical reasoning can be organised into three major phases comprised of: identification and retrieval, mapping and transference, and application (see Figure 1). These processes are described as follows:

- 1 *Identification and retrieval.* When trying to understand a design situation (i.e. the 'target'), designers represent it according to different features that may encompass abstract solution principles. These features then activate memory retrieval cues that permit access to relevant information about known or familiar situations related to past experiences (i.e. the 'source') (Gick and Holyoak, 1980; Reeves and Weisberg, 1993; Coyne and Snodgrass, 1995).
- 2 *Mapping and transfer.* A vast number of metaphorical sources can embrace a variety of potential solution principles to the design problem. In order to check the suitability of these potential sources, designers must draw structural correspondences between the selected metaphor and the design target (Sternberg and Ketron, 1982; Johnson, 1987; Vosniadou, 1989). By doing so, abstract relationships are mapped from the metaphorical source to the design situation. This is a critical stage in the process of metaphorical reasoning that demands abstractions to be made at the same level in both source and target. A successful mapping will facilitate the transference of solution principles to the design problem.

- 3 *Adaptation and application.* Having established a mapping, the use of a metaphorical principle in a design is not an automatic operation. A modification or adaptation is necessary in order to apply the solution principles to the design problem (Novick, 1988; Novick and Holyoak, 1991).

The critical role played by the relations established between source and a problem during metaphorical reasoning has been acknowledged in various studies from different research domains such as cognitive psychology (e.g. Gentner and Clement, 1988; Gentner et al., 2001) and linguistics (Tourangeau and Rips, 1991; Lakoff, 1993). However, the relation between metaphorical reasoning and learning styles was not explored. In Section 4, the potential advantages and disadvantages of the different learning styles with respect to the use of metaphor through the different phases of this process are hypothesised.

In Section 2.2, we discuss the importance of metaphors in design with particular emphasis on the design studio.

2.2 *Metaphor and design: the reflective view*

It is well acknowledged that with most design problems being vague, ill-structured, and non-routine, it is not possible to derive a solution through the simple application of tried and tested algorithms for particular problems (Goel, 1995). The use of metaphor, however, has been shown to provide a more practical strategy for such problems and as such, this approach may thus be considered an alternative to the traditional concepts fostered, in part, by the design methods movement of the 1970s (Coyne and Snodgrass, 1994).

By contributing to the redefinition and reframing of a design situation, metaphors allow the designer to perceive the design from different viewpoints. Moreover, metaphorical reasoning serves for the purposes of stimulating a reflective dialogue with the design situation (Schon, 1983b; Schon and Wiggins, 1992). According to Schon (1983b), design problem solving is characterised by a reflective conversation between the designer and his/her design materials. By identifying relevant aspects from the design problem, the designer frames a problem according to a particular situation, and develops a solution while evaluating and reflecting upon the design outcome. A contribution of the reflective conversation approach to design problem solving is that it enables designers to gain an insight on design thinking, and to verify their understanding of the design situation in particular. Through reflective practice, designers constantly construct their understanding of the design situation, and reframe it on the basis of prior experiences. Current design education research based on the analysis of design thinking and design behaviour saw in Schon's approach of design as reflection in action a potential tool for enhancing our understanding on the design process.

It is by reflective interaction that the designer can explore a large number of innovative metaphors, identify a suitable metaphorical principle, and apply it to a specific problem situation. In this way, conventional knowledge is extended to new frontiers and novel problem definitions while creative ways of generating design action emerge (Coyne, and Snodgrass, 1994). The use of metaphors is, therefore, of extreme importance as an alternative to conventional design education methods and it is for this reason that metaphorical thinking has much to contribute to design education and to the design studio in particular.

2.3 Metaphors in the design studio

In recent years, the use of metaphors in the design studio has started to capture the attention of educators and researchers, and some empirical studies have now been carried out to examine the influence of this tool in design education (mainly in the domain of architectural design). Coyne and Snodgrass (1994), for example, tested the use of metaphors in design studio practice and found that different factors (such as the historical context, the norms of practice as professionals, the conventions of the design studio, and the personal interests of teachers) determined the choice and application of certain metaphors.

In a recent empirical study carried out in the design studio, Casakin (2004b) found that metaphors help identify and retrieve design concepts as well as helping to define goals and requirements. In other research, Casakin (2006) investigated differences in the use of metaphors during different phases of the design process. It was found that metaphors were more useful in early design stages and provided more support when applied during the early stages, as opposed to the final stages, of the design process. Metaphors were thus more helpful (for novice designers at least) in the conceptual and reflective phases than in the concrete and experiential ones. In a later study that focused on the development of skills, however, Casakin (2007a) found that when students acquire more knowledge and experience, the contribution of metaphors in the final stages was improved.

Such studies, combined with the fact that the analytic power of metaphor has been shown to have a unique contribution to design innovation (Casakin, 2007b), further confirm the importance of metaphors in the design studio.

3 Metaphors and learning style

Metaphor is a useful analytic tool because it acts as a 'mental model' (Gentner and Stevens, 1983) of the problem. Interestingly, it has also been suggested that an individual's capacity to acquire and develop effective mental models is influenced by their particular 'learning style' (Wu, Dale and Bethel, 1998). It is suggested that if this relationship exists, then it might afford an opportunity to better understand and supporting the individual use of metaphor in design. As far as the relationship can be explicated, individual learning styles could be exploited, nurtured or encouraged in the design studio. Awareness of individual learning styles and their relation to metaphorical reasoning could also play a significant role in helping us understand how students with different learning abilities perceive and process metaphors in the different stages of the design process. It is therefore expected that gaining insight to this relationship will help contribute to the successful use of metaphors in the design studio. Furthermore, it is also suggested that being aware of the relation between metaphors and learning styles will aid teachers to educate design students with different learning skills in a more pluralistic way. Awareness of the relationship between metaphors and individual learning styles might foster the interactive and reflective dialogue established between the student and the design teacher. These may have consequences in three different levels: the design student, the design process and the design outcome. Regarding the design student, a result of the reflective dialogue inspired by the use of metaphors may be the identification of strengths and weaknesses of the learner. On what it concerns the design process, the

reflective dialogue may contribute to an incremental development of design solutions following the different stages of the metaphorical process. This in turn, will lead to the production of creative and innovative design outcomes.

3.1 Learning styles

The concept of learning style is, however, not particularly well understood – indeed, there is no one definition of it in literature. It is known variously as ‘thinking styles’, ‘cognitive styles’ and ‘learning modalities’ and there are as many different theories as there are terms for the idea. Unfortunately, reliable studies do not so easily accompany each of these theories, and the lack of consensus can be directly attributed to this paucity of evidence.

Working from the studies that have been made however, some researchers has broadly commented on the field in general, identifying characteristics that should be borne in mind (BECTA, 2005):

- Any theory or model is necessarily a simplification.
- Learning styles are at best one in a range of factors that determine learning.
- It is at least as effective to match the presentation type with the nature of the subject matter as it is to match it to individual learning styles.
- Awareness of their learning style may help students to develop a metacognition of how to learn.

With these caveats in place, it is worth briefly discussing the learning styles that have emerged in the literature.

3.2 Learning theories

As mentioned, it is possible to find various studies of learning styles in the literature, from those with a basis in behaviourist, Gestalt or Jungian psychology, to others that emphasise the role of the senses or that have evolved from popular notions of neurology or brain structure (Cassidy, 2004). There are, however, four models of particular note due to their prevalent application:

- 1 The Myers–Briggs type indicator (MBTI) classifies learners according to their preferences on scales derived from Carl Jung’s theory of psychological types. MBTI measures, among other things, the degree to which an individual prefers sensing or intuition (Wankat and Oreovicz, 1993).

The characteristics of intuitive and sensing types, and the different ways in which sensors and ‘intuitors’ approach learning have been studied and are articulated as four bi-polar scales of (Felder and Silverman, 1988):

- perceiving/judging
- sensing/intuition
- thinking/feeling
- extraversion/introversion.

Being based on Jung's psychological types, this approach is not just used to categorise learning, but also personality. As such, in this model, people may be classified as extraverts or introverts, sensors or intuitors, thinkers or feelers and judgers or perceivers.

- 2 The Felder–Silverman model of learning styles incorporates five dimensions (some incorporated from MBTI and Kolb). The principle dimensions and relations are as follows (Felder and Silverman, 1988):

- active and reflective learners
- sensing and intuitive learners
- visual and verbal learners
- inductive and deductive learners
- sequential and global learners.

The conceptual framework developed by Felder and Silverman therefore allows for 32 individual learning styles in all, based on different combinations of these dimensions.

- 3 Herman brain dominance instrument (HBDI) uses the (supposed) task-specialised functioning of the quadrants of the physical brain to classify people according to their particular preferences (Cassidy, 2004). Thus:

- theorists (left brain, cerebral)
- organisers (left brain, limbic)
- humanitarians (right brain, limbic)
- innovators (right brain, cerebral).

While classifying individuals into these various groups, the approach does emphasise the need for people to develop a versatility that will allow them to learn and perform in a variety of different situations.

- 4 Kolb's ELT suggests learning to be a dynamic and cyclical process (Stumpf and McDonnell, 2002). It considers the idea of reflection to be fundamental to our ability to learn – indeed, it holds that without reflection, we would simply continue to repeat our mistakes.

Building on existing experiential theory, Kolb (1984) conceived of learning as a circular process, and by refining the concept of reflection, separated learning into the two separate processes of receiving and internalising information (Demirbas, 2001). By considering these processes as two orthogonal continuums (axes) within this circular process, the extremes these can be taken to define four stationary points in the learning process – or the four modes of learning. These are:

- concrete experience – putting into practice
- observation and reflection – objective analysis
- abstract conceptualisation – reviewing conceptual understanding
- active experimentation – experimenting to find solutions.

The result of this is what is known as the experiential learning model. According to this model, concrete experience is followed by observation and reflection which then leads to the development of abstract concepts and generalisations which are then tested through active experimentation (Demirbas, 2001).

While an individual's preferences for receiving and internalising information is considered to determine their particular learning style, it should be noted that the theory places much emphasis on development in each mode and encourages individuals to become competent in all styles in order to be a balanced, integrated learner.

3.2.1 Adopting Kolb's experiential learning theory

While acknowledging the caveats associated with any learning styles in general (cf. Section 3.1), there is much to commend Kolb's ELT as a framework for the study proposed in this article.

First of all, previous application of ELT in an increasing corpus of studies (including in design literature) as well as the suggestion of the influence of experience in the metaphoric design process (cf. Section 2.3) suggest it might provide new insights.

Furthermore, by having no commitment to personality types or questionable brain theory, ELT does not consider learning style to be innate and unchangeable. Moreover, by adopting a holistic perspective – combining 'experience, perception, cognition and behaviour' (Kolb, 1984) – it raises awareness of various aspects of learning that are not considered by other approaches. Finally and importantly, by conceptualising learning as a cycle, it also helps students develop a 'metacognition' of learning (cf. Section 3.1), encouraging them to become more aware of their position in the cycle and to build strengths in areas of weakness.

3.3 Experiential learning

ELT, being based on the work of individuals, such as Piaget, Dewey and Merleau-Ponty, focuses on immediate experience and places the idea of situatedness at the centre of learning (Dewey, 1938; Flavell, 1963). While it has been developed from theories built as a response to more traditional, behavioural theories of learning, it should be noted that it was never intended as an alternative to cognitive or behaviourist theories. Rather, it was developed to provide a 'holistic, integrative perspective on learning that combines experience, perception, cognition and behaviour' (Kolb, 1984). Kolb defines several characteristics of the theory:

- Learning is best conceived of as a process, not in terms of outcomes – ideas are formed and reformed through experience.
- Learning is a continual process grounded in experience – learning emerges from the interplay between expectation and experience.
- Learning involves transactions between the person and the world (transaction being distinct from the limited concept of interaction).
- Learning requires the resolution of conflicts between dialectically opposed models of adaptation to the world.

Where other theories emphasise the acquisition, manipulation and recall of abstract symbols (Kolb, 1984), experiential theory argues for the intrinsic role of experience in giving meaning to symbols, the concepts they represent and their potential interpretation and use.

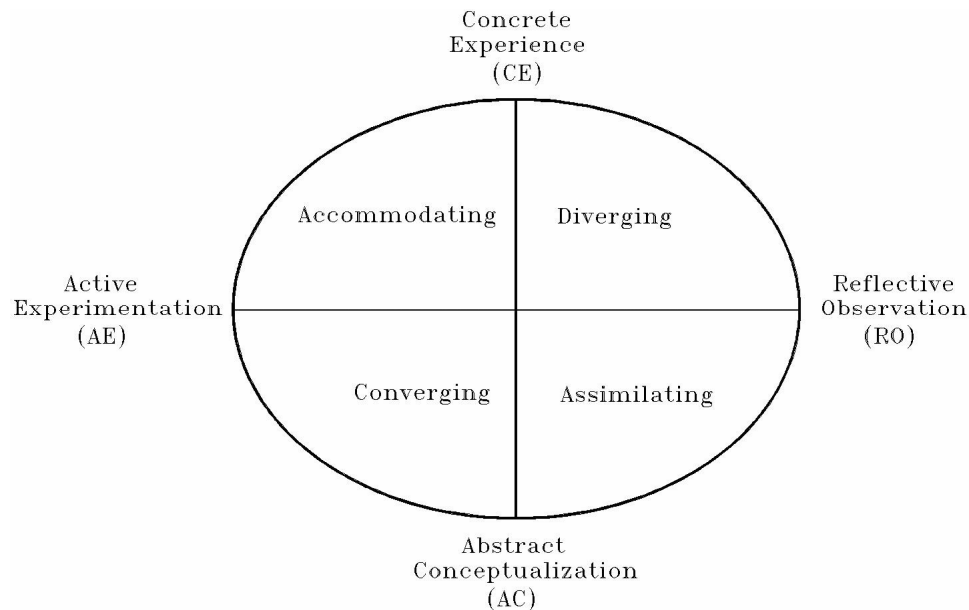
This is particularly important when considered in terms of the mental models we have developed and how they are used. As has been observed (Bodner, 1986), many students find abstract, mathematical arguments and models insufficient reason to reject the naïve (but intuitive and, importantly, meaningful) models they have developed. Indeed, such resistance has been shown to cause the retention and misapplication of pre-Newtonian to problems amongst engineering students (Svanaes, 1997; Dym et al., 2005). Experience is thus considered to play a key role not only in the models we develop, but also in the way they are applied to new situations.

3.3.1 *The experiential learning cycle*

Kolb argues that learning styles are not determined by inherited characteristics, but develop through experience (Kolb, 1984). He suggests that it is the combination of how people perceive and how they process information that forms the uniqueness of their own learning style, and has articulated this process in a four-stage model comprised of the two continuums (see Figure 2).

The concrete-abstract continuum (vertical axis) is concerned with the way we perceive new information. The theory suggests that when presented with a novel situation, those close to concrete experience rely on intuition and prefer to sense their way through it, while those close to abstract conceptualisation are inclined to distance themselves so that they can consider and analyse the situation.

Figure 2 The experiential learning cycle as proposed by Kolb



The active-reflective continuum (horizontal axis) is concerned with how we process new information. Some people might have a tendency to jump in and try things out (those close to active experimentation) while others prefer to contemplate and assess (those close to reflective observation).

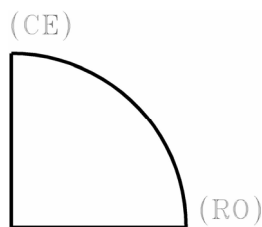
According to the theory, the extremes of each continuum are mutually exclusive. If, for example, we try to perceive new information by concrete experience and by abstract conceptualisation, a conflict occurs (Wu, Dale and Bethel, 1998). Each individual must therefore make a choice in order to resolve the conflict, and it is in this way that each individual develops a preference, that is a learning style, for the perception and processing of new information.

It is important to recognise that Kolb (1984) considers this to be a model of a learning cycle – the extremes of each continuum being steps in a learning process. While most of us will usually start with concrete experience, the cycle may be entered into at any stage. Also, note that no one stage of the cycle is better than another, this means that an individual's preferences do not make them better or worse learners – although, as mentioned (cf. Section 3.2) a proficient learner will be able to complete all the steps in the cycle (despite having a preference for certain modes of operation).

3.3.2 *Experiential learning styles*

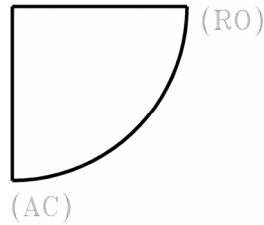
By considering the continuums as bipolar axes, individual preferences can be thought of as coordinates that can be used to identify the position of an individual within the model. Therefore, in a further development of the theory, each quadrant has been identified with a particular learning style (or preference):

Diverging learners. A divergent learner's principle preferences are for concrete experience and reflective observation. Individuals with this learning style are most proficient at collecting and synthesising a wide variety of information and, as such, are adept at viewing concrete situations from a variety of perspectives. Being less interested in theory and apt to deal with situations in a non-systematic way, they are not very good at taking decisions. They tend to be emotional, imaginative and enjoy working in groups and perform best in situations calling for the generation of innovative ideas (when, for example, using 'creative thinking' methods, like brainstorming).

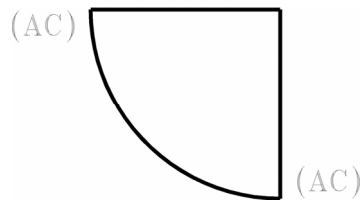


Assimilating learners. An assimilating learner's principle preferences are for abstract conceptualisation and reflective observation and, as such, cannot only understand a broad range of information, but are also capable of organising the information into a succinct and coherent form. Assimilators are less interested in people and are more concerned with ideas and concepts – and value the logic and precision of these ideas rather than

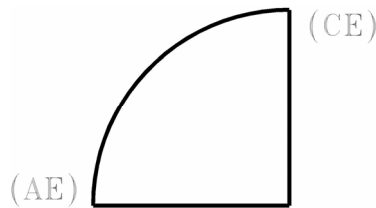
their practical value. As such, people with this learning style like to take the time to think things through and prefer to read and carefully explore models.



Converging learners. A converging learner has the opposite abilities of a diverging learner and therefore has principle preferences for abstract conceptualisation and active experimentation. They are logical, organised and pragmatic thinkers who, when solving problems and making decisions, excel at finding practical applications for theories and ideas. They are not emotional and prefer to deal with things rather than people – focusing their knowledge on specific problems and hypothetical-deductive reasoning.



Accommodating learners. Having the opposite characteristics of assimilators, the principle preferences of accommodators are for concrete experience and active experimentation. Their learning relies heavily on experience and so enjoy taking risks and constantly challenging themselves. Rather than following a scientific or systematic approach to a problem, they prefer to act according to their instincts and intuitions and this will solve problems through trial and error. They will also tend to eschew their own analytic ability, opting instead to get information from others.



4 Proposed hypotheses

While no particular learning style is intrinsically superior to any other, there might be some differences in the likely performance of individuals with particular learning styles with regard to the metaphorical reasoning processes described earlier. Not only might

such considerations be of importance to design, but also it is worth noting that the use of metaphors is an issue not yet fully addressed by Kolb's ELT.

With respect to the different learning styles, then, areas of possible advantage and disadvantage have been suggested in Table 1 displayed below:

Table 1 Performance of designers in the different phases of the metaphorical process according to their individual learning styles

Learning style	<i>Phases in the metaphorical process</i>					
	<i>Identification and retrieval</i>		<i>Mapping and transfer</i>		<i>Adaptation and application</i>	
	Inf. Per.	Inf. Pro.	Inf. Per.	Inf. Pro.	Inf. Per.	Inf. Pro.
Convergers	A	D	A	A	A	A
Divergers	D	A	D	A	D	D
Accommodators	D	D	D	D	D	A
Assimilators	A	A	A	A	A	D

Inf.Per., information perception; Inf.Pro., information processing; A, advantage; B, disadvantage.

By elaborating on the possible advantages and disadvantages of the different learning styles during the use of metaphor, it might be possible to better understand how to train students in the design studio. Therefore, we propose an analysis of potential relations between learning styles and the different phases of metaphor use in design problem solving, and present hypotheses on the contribution of learning styles in each phase of the metaphorical process.

1 Identification and retrieval

As mentioned earlier, the first phase of the metaphorical process is concerned with the identification of metaphorical sources and the retrieval of metaphorical principles. At this stage, designers process and reflect on information related to the design situation from new viewpoints. While doing this, they also start looking for candidate metaphorical sources that are familiar to them, but remote or outside the design realm. This is then accessed by means of abstract principles that act as retrieval cues.

Hypothesis 1A. Assimilators will perform best in the identification and retrieval phase.

Hypothesis 1B. Accommodators will perform worst in the identification and retrieval phase.

- *Assimilating* learners, who combine abstract conceptualisation and reflective observation, will be most successful in this phase of the process.
- *Converging* learners, who combine active experimentation with abstract conceptualisation, are also expected to succeed in retrieving abstract concepts from the metaphorical source. However, they are also expected to fail to reflect on the nature of the situation from a new perspective. It is possible that the retrieved principle will not be novel enough, and as a result, the solution to the problem will be quite familiar.

- *Diverging* learners, who are opposite to converging learners, despite being able to reflect on the nature of the design situation from unorthodox and innovative perspectives, will be impeded to reason in terms of abstract ideas and will thus have difficulties retrieving successful metaphorical principles from the source.

We postulate that ‘accommodators’, who perceive information in concrete terms and have a strong preference for active experience, will be the most disadvantaged learners at this stage. These individuals will not only have difficulty reflecting on the nature of the design situation anew, but they will also be inefficient at making abstractions and retrieving concepts from the metaphorical sources.

2 *Mapping and transfer*

The second phase of the metaphorical process is characterised by establishing structural correspondences between the metaphorical source and the design problem. At this stage, designers need to figure out how the principle retrieved from the metaphorical source can be mapped and transferred to the problem at hand. This means establishing a structural alignment between two different situations and then projecting inferences through a mapping of abstract relationships.

We thus hypothesise that learners who will be successful at this stage will be those who are able to perceive information in terms of abstract conceptualisations. We also maintain that depending on the learning style of the designer, correspondences between the source and design problem might be established either by active experimentation or by reflective observation.

Hypothesis 2A. Converging and assimilating learners be most successful at establishing abstract correspondences.

Hypothesis 2B. Assimilators will develop the most novel ideas from remote domains.

Hypothesis 2C. Diverging and accommodating learners will have most difficulty establishing abstract correspondences.

With this in mind then, we hypothesise that the most successful designers in this phase will be converging and assimilating learners. Both are expected to be effective at abstraction and establishing correspondences between source and target. However, we do stress, that the quality of the metaphorical principle transferred to the problem will vary in each case.

Since converging learners will tend to process information by active experimentation within the design domain (i.e. by trial and error), it is possible that the outcome will not differ largely from familiar design solutions. In contrast, assimilators, when reflecting about the nature of the correspondences, will search for novel ideas and will therefore attempt to establish remote and high-quality alignments between source and target.

We suggest that accommodators and diverging learners will be the most disadvantaged at this stage. Diverging learners will be interested in processing information by reflecting on it. However, since they perceive information through concrete experience, making abstractions and establishing mappings is expected to be awkward for them. Accommodators on the other hand, will process information by actively experiencing the design problem and will learn by doing throughout the design task. Therefore, they are expected to fail to reflect from a novel viewpoint on the nature

of the design problem and to have difficulty making abstractions and establishing structural correspondences.

3 *Adaptation and application*

The third phase of the metaphorical process is concerned with the adaptation of the metaphorical principle to the target situation. After projecting inferences through a mapping of relationships, designers should apply an abstract concept to a concrete problem situation. We postulate that at this stage, successful designers will have to make abstract conceptualisations in order to adapt the metaphorical principle to the design situation, and will have to actively experiment to develop it into a concrete solution.

Hypothesis 3A. Converging learners are expected to be best at adapting and applying metaphorical principles to the design situation.

Hypothesis 3B. Assimilators will be able to organise the metaphorical idea, but will have most difficulty in applying metaphorical principles to the design situation.

Consequently, we hypothesise that converging learners, who are pragmatic thinkers and combine both abstract conceptualisation and active experimentation will perform better in this stage of the process. Accommodators are expected to prefer to act according to their own feelings instead of considering a logical or analytical approach and will, therefore, succeed in developing a solution, but not one of particularly high-quality or innovation.

In contrast, assimilators will manage to organise the metaphorical idea in a concise and logical form, but lacking the abilities to recognise practical values of an idea, will find it difficult to apply to a concrete design problem. There are good chances that they will produce a solution embracing a creative principle, but the outcome will be very schematic with little elaboration or detail.

Finally, we posit that diverging learners, who are best at dealing with concrete situations, will have difficulty thinking in terms of abstract ideas and in adapting them to a design problem. This will reduce their abilities and will impair their development of quality design solutions.

5 **Discussion and conclusions**

In this article, we have discussed the role played by metaphorical thinking in design with particular focus on the design studio and education. The component processes of metaphorical reasoning have been presented, and the well established Kolb's experiential learning models, as well as its four classes of learning style, have been described. The ELT was further analysed with regard to design and to the use of metaphors in design before the main hypotheses about the effect of different learning styles on each phase of the metaphorical process were proposed.

Instead of attempting to offer generalisable rules or clear cut conclusions about the use of metaphors in design problem solving, the present study tried to explore and discuss some assumptions concerned with the use of this tool in the design studio by students with different learning styles. By aiming to establish the nature of the relationship between learning styles and metaphor, it is expected that the work proposed here will provide insight to student performance in design and, in so doing, also gain understanding about strengths and weaknesses of each learning style and the way in

which each could serve as predictors of design success or failure in the metaphorical reasoning process.

While bearing in mind the caveats already discussed (e.g. that any model of learning is by necessity a simplification), it may be instructive to identify general learning styles of students who are comfortable with metaphor use in order to learn from their approaches to support other students and move them through the learning cycle. It is also hoped that such a study will help raise students' awareness of their own learning styles in order that they might consider their approach more carefully, develop their skills in the design studio, and ultimately learn how to learn.

5.1 Hypotheses

Generally speaking, it has been proposed that assimilators, followed by converging learners, hold learning preferences most advantageous for the successful use of metaphor. Assimilators are characterised by abstract conceptualisation and reflective observation abilities which are expected to be useful in most metaphorical phases. In order to fully succeed, however, it is supposed that assimilators will need training in the last phases of the design process in order to develop skills for the application of metaphorical knowledge to concrete design situations. Converging learners, on the other hand, are expected to require assistance to develop reflective observation skills in the earlier phases of the process.

It is suggested that diverging learners and accommodators are considered to be the most disadvantaged learners (respectively) with regard to metaphorical reasoning. Both of them are expected to need training to perceive and integrate information by means of abstract conceptualisation which is essential to the earlier stages of the metaphorical process. It is also suggested that accommodating learners, who tend to act on intuition, will need assistance when analysing information from different perspectives and will have to be encouraged to reflect on information before taking action.

5.2 Implications

What could be the implications for design education? It is suggested that in light of the importance of metaphor in design practice, the training of students in the use of metaphor is by itself a valuable accomplishment. Reflecting about design problems through metaphor may enable designers to enhance their design thinking abilities, and their understanding of the design problem. The reflective practice will provide them arguments and concepts from unconventional perspectives that may lead to innovative designs. Students, who may come to a situation with entrenched preconceptions, are therefore encouraged to assume a more reflective attitude towards the problem and to engage in a dialogue to allow suitable metaphorical ideas to emerge.

In addition, the analysis carried out in this work concerning the effect of learning styles on metaphorical reasoning has provided insight on a number of aspects. First, the use of metaphors is a process composed of a series of consecutive phases, each of which have different requirements. Second, no single learning style appears to have been consistently strong across all the phases of this process – each of them seem to have both advantages and disadvantages that vary with the different requirements.

Awareness of the differences of learning styles might provide studio teachers with a refined tool to train students in metaphor use. Furthermore, it will encourage teachers to

educate design students with different learning skills in a more pluralistic way. In this way, it will be possible for teachers to exploit the advantages of each learning style regarding each phase of the metaphorical process and thus help students in those phases that are weaker.

5.3 Future work

In a future study, we aim to empirically test the theoretical postulates presented in this article. We plan to provide design students with a design task and explicitly ask them to approach the situation using metaphorical reasoning in order to assess performance in each phase of the process. In a second stage, we intend to use the learning style inventory developed by Kolb to determine the learning style of each individual student and thus attempt to assess its effect on metaphorical reasoning.

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